SHIFT LEVER (Continued)

(6) Remove screws attaching lever assembly to floorpan and remove assembly and shift rod (if left attached).

INSTALLATION

(1) If shift rod was not removed from lever assembly, work rod down through floorpan opening. Then position lever assembly on floorpan and install assembly attaching screws.

(2) Install console. (Refer to 23 - BODY/INTERIOR/FLOOR CONSOLE - INSTALLATION)

(3) Raise vehicle.

(4) Connect trunnion to torque shaft arm. Or, slide shift rod into trunnion on range lever. Be sure shift rod slides freely in trunnion.

(5) Verify that range lever is in 4L position. Then tighten trunnion lock bolt.

(6) Lower vehicle and check transfer case shift operation.

ADJUSTMENTS - SHIFT LINKAGE

(1) Shift transfer case into 4L position.

(2) Raise vehicle.

(3) Loosen lock bolt on adjusting trunnion (Fig. 103).

(4) Be sure linkage rod slides freely in trunnion. Clean rod and apply spray lube if necessary.

(5) Verify that transfer case range lever is fully engaged in 4L position.

(6) Tighten adjusting trunnion lock bolt.

(7) Lower vehicle.

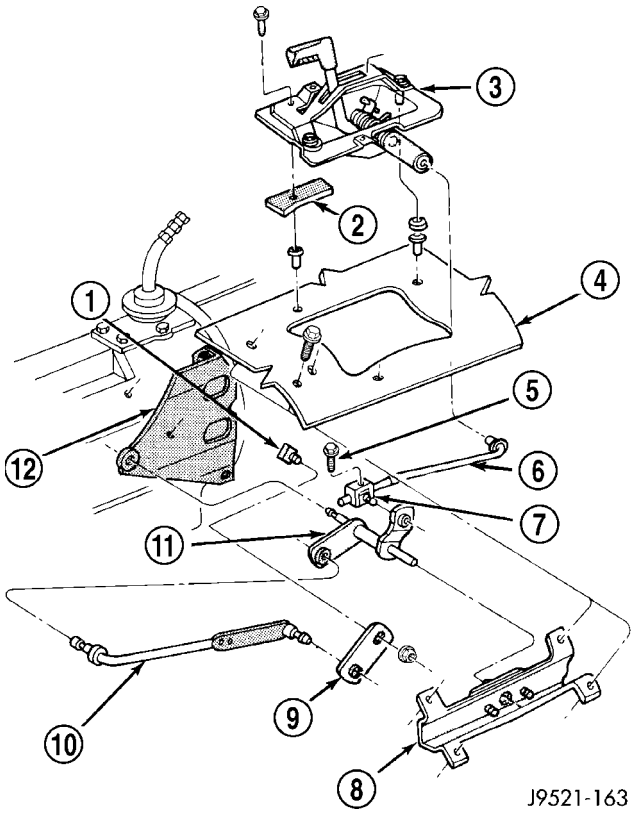

Fig. 103 Shift Linkage

1 - TRANSFER CASE SHIFT LEVER SHAFT
2 - SEAL
3 - TRANSFER CASE SHIFT LEVER ASSEMBLY
4 - FLOORPAN
5 - TRUNNION LOCK BOLT
6 - SHIFT ROD
7 - ADJUSTING TRUNNION
8 - TORQUE SHAFT BRACKET
9 - RANGE LEVER
10 - TORQUE SHAFT ROD
11 - TORQUE SHAFT
12 - LINKAGE BRACKET

VEHICLE SPEED SENSOR

DESCRIPTION

The 3-wire Vehicle Speed Sensor (VSS) (Fig. 104) is located in the transfer case rear extension.

OPERATION

The VSS is a 3-circuit (3–wire), magnetic, hall-effect sensor.

The 3 circuits are:

- A 5-volt power supply from the Powertrain Control Module (PCM).
- A ground is provided for the sensor though a low-noise sensor return circuit in the PCM.
- An input to the PCM is used to determine vehicle speed and distance traveled.

The speed sensor generates 8 pulses per sensor revolution. These signals, in conjunction with a closed throttle signal from the throttle position sensor, indicate a closed throttle deceleration to the PCM. When the vehicle is stopped at idle, a closed throttle signal is received by the PCM (but a speed sensor signal is not received).

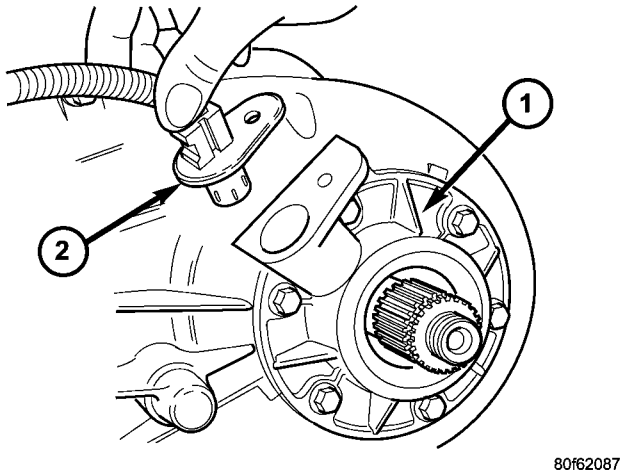

Fig. 104 Speedometer Sensor

1 - EXTENSION HOUSING
2 - SPEEDOMETER SENSOR